

More field-programmable gate arrays in DAQ system

Field-programmable gate arrays (FPGAs) have made their way to DAQ systems for in-line signal processing and data reduction. This has resulted in more intelligent DAQ systems, which you can program with various available software tools to process acquired data on the device without having to transfer it over a bus to the PC processor. When it comes to high-end measurement applications involving immediate processing on the incoming data, on-board FPGAs come into play.

Also, the software tools for these programmable solutions with pre-defined as well as custom algorithms are leveraging the FPGA technology. This enables user-defined control, sensor simulation, digital protocol emulation with flexible and precise timing and triggering, allowing you to configure hardware and make deterministic measurements and control.

Owing to their intelligence and flexibility, these devices can be ideal solutions for custom DAQ requirements, embedded applications and development and testing of custom digital interfacing. As technology advances, the price/performance evolution curve of FPGAs will continuously improve, and also with various software and tools easing the programming of FPGAs, we can expect the integration of FPGAs on DAQ systems to continue to expand.

DAS becomes more affordable

As far as a measurement system is concerned, we cannot say that the prices decrease sharply. But due to a growing processing capability, you can get more processing and measurement capability available for the same amount of money you spent before.

Following the trend of other semiconductor technologies, modern-day ADCs have significantly reduced costs. Today, we have advanced integrated ADCs with faster conversion speeds, higher resolutions, at an order of much lower price than what was previously

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available.

The price decline has contributed to the integration of signal conditioning features on DAQ systems for the improvement of safety, accuracy and ease of use. ADCs with 16-, 12- or even 10- and 8-bit resolution are available depending upon what requirement you are looking for.

A very important thing that has to be kept in mind is, a lot of instrument functionality controlled by the software that you use to control these equipment, and with the PCs becoming affordable and powerful, you are able to use the equipment to a much more extent than ever before. So, as a user, your overall cost for DAQ still reduces because PC still owns a larger part of your system and you are getting much more value for the money you spent.

Frankly speaking, you will not see a drastic drop in DAQ systems' cost like you see in consumer electronics. However, if we see overall value for money, we can see a drastic change in terms of processing, power and measurement capability. Vendors such as Measurement Computing Corporation have lowered the cost of DAQ by bringing out inexpensive USB-based versions that can sample at 500kS per second for under US\$ 200.

Future

We can see that the technology advancements in the broader computing and consumer electronics sector will continue to leverage and benefit the future DAQ systems. We can expect to

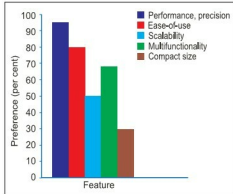


Fig. 3: Most preferred DAS features (Source: Integrated Process Systems)

see continued growth and adoption of USB, more integration of signal conditioning and added flexibility with on-board programmable FPGAs on DAQ systems. With increased modularity and granularity, the next-generation interfacing and integrated wireless capabilities, one can look forward to the development of more advanced DAQ in the near future.

Another area which most of the companies are eyeing for future is Cloud-based DAQ. The applications are already developed and available for various popular operating systems, such as Android and iOS, for remote DAQ with handheld devices allowing you to see the sensor data and providing you the ability to control various parameters, and also log the data onto the system. Working in a seamless environment provides users the ability to access their data from anywhere, further increasing the flexibility and utility of the systems. ●

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